

APPENDIX T1:

Surveys may be administered via paper-based surveys distributed to occupants or through a digital survey such as a website or mobile application. It is recommended that surveys are administered during mid-morning or mid-afternoon, at least 30 minutes after arrival or after a lunch break.

Survey Questionnaire:

Please answer the following questions regarding your general thermal experience during the current heating/cooling season.

1. Please rate your satisfaction level with the thermal environment in this space:

- ☐ Very Dissatisfied
- ☐ Dissatisfied
- ☐ Slightly Dissatisfied
- ☐ Neutral
- ☐ Slightly Satisfied
- ☐ Satisfied
- ☐ Very Satisfied

2. Would you prefer this area to be:

- ☐ Cooler
- ☐ Slightly Cooler
- ☐ No Change Required
- ☐ Slightly Warmer
- ☐ Warmer

If your answer to Question 1 contains the word "Dissatisfied," please answer the following two questions:

3. How do you generally perceive the thermal environment in this area:

- ☐ Cold
- ☐ Cool
- ☐ Slightly Cool
- ☐ Neutral
- ☐ Slightly Warm
- ☐ Warm
- ☐ Hot

4. Please mark the potential source (or sources) of your thermal discomfort:

- ☐ Inappropriate room thermostat setpoint temperature
- ☐ Thermostat setpoint temperature being adjusted by occupants
- ☐ Daily variations in room temperature (such as higher temperature in the afternoons)
- ☐ Occasional variations in room temperature (such as being occasionally warm or cold)
- ☐ Parts of the room being too hot
- ☐ Parts of the room being too cold
- ☐ Humidity level is too high
- ☐ Dry air
- ☐ Air movement is too high

- ☐ Air movement is too low
- ☐ Hot/cold air coming from windows
- ☐ Hot/cold walls
- ☐ Solar radiation
- ☐ Local thermal discomfort (part/parts of the body being cold or hot)
- ☐ Heat from equipment and appliances
- ☐ Strict dress code
- ☐ Hot/cold furniture surfaces
- ☐ Furniture causing hot/cold thermal discomfort or sweating
- ☐ Others

Example of Result Analysis:

The Percentage of Satisfaction should be reported as the percentage of people who are satisfied with the thermal environment of the space as shown in the following equation:

$$\text{Percentage of Satisfaction} = (\text{Number of Satisfied Occupants}) / (\text{Number of Surveyed Occupants}) \cdot 100$$

Occupants voting for "Neutral," "Slightly Satisfied," "Satisfied," and "Very Satisfied" should be deemed satisfied with the room's thermal environment. Table 1 shows the results from a typical survey and Table 2 depicts a simple example of the survey report. The results of question 2 (and question 3 if applicable) should be submitted as the number of votes for each thermal sensation and preferences condition as specified in Table 2.

Table 1: Results from a Typical Thermal Comfort Satisfaction Survey with 15 Participants

	Question 1	Question 2	Question 3	Question 4
Subject 1	Very Satisfied	No Change		
Subject 2	Satisfied	Slightly Cooler		
Subject 3	Slightly Dissatisfied	Warmer	Cold	Parts of the room being too cold
Subject 4	Slightly Dissatisfied	Slightly Warmer	Cool	Hot/Cold air coming from windows
Subject 5	Satisfied	No Change		
Subject 6	Very Satisfied	No Change		
Subject 7	Neutral	Slightly Warmer		
Subject 8	Very Satisfied	No Change		
Subject 9	Dissatisfied	Warmer	Slightly Cool	Inappropriate room thermostat setpoint temperature
Subject 10	Slightly Satisfied	Slightly Cooler		
Subject 11	Neutral	Slightly Warmer	Slightly Cool	Hot/Cold air coming from windows
Subject 12	Slightly Satisfied	Slightly Warmer		
Subject 13	Very Satisfied	No Change		
Subject 14	Slightly Satisfied	Slightly Warmer		
Subject 15	Satisfied	No Change		

Table 2: A typical Thermal Comfort Satisfaction Survey Report

1. Please rate your satisfaction level with the temperature in this room:

Considered Dissatisfied			Considered Satisfied			
Very Dissatisfied	Dissatisfied	Slightly Dissatisfied	Neutral	Slightly Satisfied	Satisfied	Very Satisfied
0	1	2	2	3	3	4

Percentage of Occupant Satisfaction:

80%

2. Would you prefer this area to be (based on the number of votes):

Cooler	Slightly Cooler	No Change	Slightly Warmer	Warmer
0	2	6	5	2

3. How do you generally perceive the thermal environment in this area (based on the number of votes):

Cold	Cool	Slightly Cool	Neutral	Slightly warm	Warm	Hot
1	1	2	0	0	0	0